**MBA652 - STATISTICAL MODELLING FOR BUSINESS ANALYTICS**

**DEPARTMENT OF MANAGEMENT SCIENCE (DOMS)**
**INDIAN INSTITUTE OF TECHNOLOGY KANPUR**

# PROJECT: Analysing Factors Affecting Nitrogen Oxides (NOx) Emissions of Gas Turbine

## Submitted to –

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**OBJECTIVE:**

The aim of our project is to analyze operational and environmental factors affecting the Nitrogen Oxides (NOx) emissions of an industrial gas turbine and develop a multiple linear regression model to predict emission levels for environmental compliance and process optimization.

**BACKGROUND:**

Industrial gas turbines operate under continuously varying ambient conditions and load demands, where peak combustion flame temperatures trigger thermal NOx formation—a toxic atmospheric pollutant subject to strict statutory emissions ceilings. Accurate estimation of NOx emissions from standard supervisory sensor telemetry (pressures, temperatures, power output, and ambient conditions) is vital for industrial power plants. It enables continuous verification of environmental compliance, detects combustion anomalies, and provides a dependable computational proxy whenever physical Continuous Emission Monitoring Systems (CEMS) undergo maintenance or sensor failure.

**METHODOLOGY:**

We have used the following models for the Project –

- Multiple Linear Regression Model (OLS with backward elimination and iterative VIF-based multicollinearity reduction)
- Heteroskedasticity diagnostic testing (Breusch-Pagan Test) and HC3-Robust Standard Error inference

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**DATA SOURCE:**

1. Dataset – Gas Turbine CO and NOx Emission Data Set (2011–2015)

https://archive.ics.uci.edu/ml/datasets/Gas+Turbine+CO+and+NOx+Emission+Data+Set

**Source:**

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Department of Computer Engineering, Çorlu Faculty of Engineering
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**Data Set Information:**

Sample Size: 36,733 hourly-averaged raw sensor records across 5 years (2011–2015); cleaned sample of 35,347 complete observations post standardized z-score outlier removal ($|z| \leq 3$).
Source: Operating industrial cogeneration gas turbine (north-western Turkey) equipped with SCADA data logging.

**VARIABLES:**

**Input Variables:**

x1 – Ambient Temperature (AT)

x2 – Ambient Pressure (AP)

x3 – Ambient Humidity (AH)

x4 – Air Filter Differential Pressure (AFDP)

x5 – Gas Turbine Exhaust Pressure (GTEP)

x6 – Turbine Inlet Temperature (TIT)

x7 – Turbine After Temperature (TAT)

x8 – Turbine Energy Yield (TEY)

x9 – Compressor Discharge Pressure (CDP)

**Output Variable:**

y = Nitrogen Oxides Emissions (NOx)

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**VARIABLE DESCRIPTION:**

## Dependent Variable

Variable which has to be forecasted and estimated. The value of this variable depends on operational and atmospheric parameters.

- **NOx (Nitrogen Oxides Emissions):** The flue gas concentration of nitrogen oxides emitted into the atmosphere, measured in milligrams per cubic meter ($mg/m^3$).

## Independent Variables

The operational and atmospheric variables which affect the dependent variable. A total of 9 independent variables have been considered in our initial model.

1. **Ambient Temperature (AT) –** The ambient air temperature surrounding the intake measured in degrees Celsius (°C), which directly alters air density and mass flow rate.
2. **Ambient Pressure (AP) –** The local barometric atmospheric pressure in millibars (mbar), affecting intake air density and compressor expansion pressure.
3. **Ambient Humidity (AH) –** The percentage relative humidity of ambient air (%), which acts as a thermal heat sink/diluent during combustion.
4. **Air Filter Differential Pressure (AFDP) –** The pressure drop across the intake air filtration system in millibars (mbar), indexing filter restriction and airflow drag.

5. **Gas Turbine Exhaust Pressure (GTEP) –** The flue gas exhaust back-pressure measured exiting the turbine expander in millibars (mbar).
6. **Turbine Inlet Temperature (TIT) –** The hot gas temperature entering the turbine section from the combustion chamber in degrees Celsius (°C), governing peak flame temperature and thermal NOx formation.
7. **Turbine After Temperature (TAT) –** The exhaust gas temperature measured downstream of the turbine expansion blading in degrees Celsius (°C).
8. **Turbine Energy Yield (TEY) –** The generated net electric power output of the gas turbine generator in megawatt-hours (MWh).
9. **Compressor Discharge Pressure (CDP) –** The air pressure leaving the axial compressor before entering the combustion cans in millibars (mbar).

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**Scatter plots:**

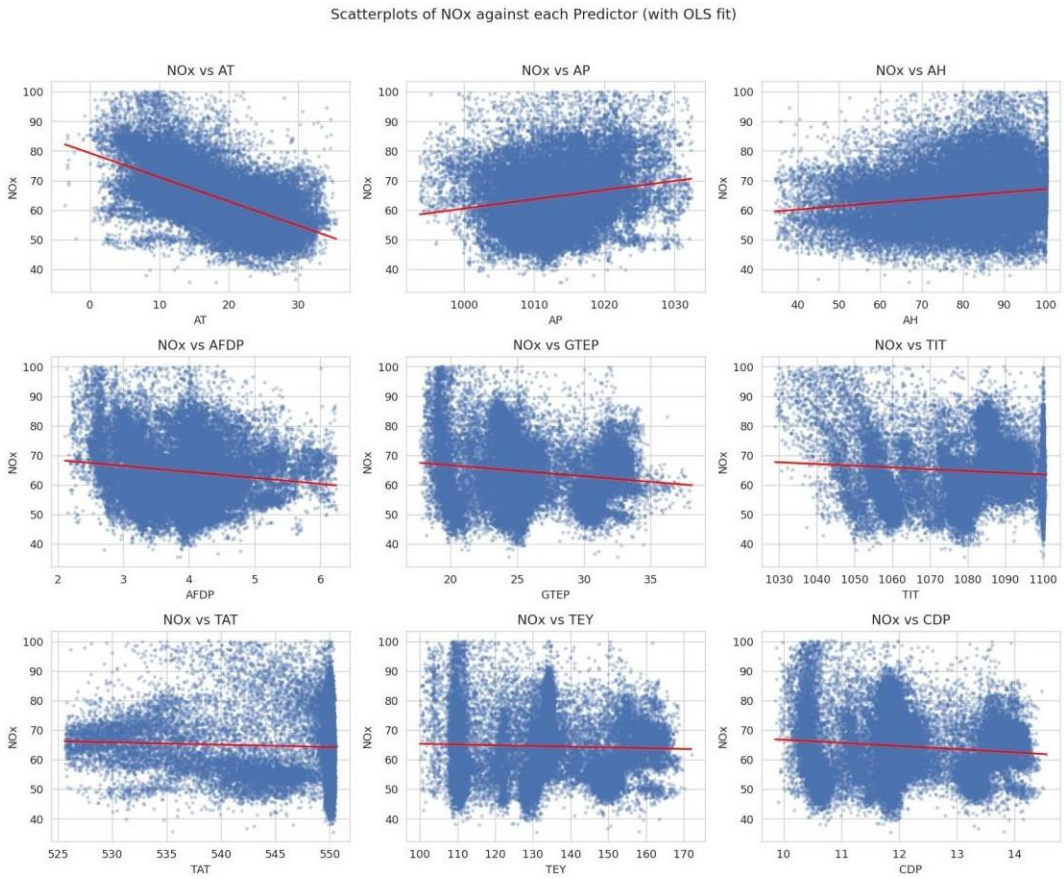



*Figure 1: Scatter plots of NOx vs. Independent Variables with Fitted Regression Lines (scatter_plots.png)*

With the help of scatter plots, the trend in the variables was analysed. It gives a rough idea of the type of correlation (positive, negative or zero) and whether the distribution of variance is heteroskedastic or homoskedastic.

• AT exhibits the clearest downward-sloping linear trend (negative correlation), indicating that higher ambient temperatures suppress NOx.

• AP and AH show weak positive correlations.

• AFDP, GTEP, TIT, TAT, TEY, and CDP exhibit weak negative bivariate slopes characterized by discrete, clustered multi-modal bands, reflecting distinct operational load regimes rather than continuous dispersion.

**Correlation Matrix:**

| | AT | AP | AH | AFDP | GTEP | TIT | TAT | TEY | CDP |
|---|---|---|---|---|---|---|---|---|---|
| **AT** | 1.00 | -0.39 | -0.48 | 0.27 | 0.06 | 0.17 | 0.23 | -0.09 | 0.03 |
| **AP** | -0.39 | 1.00 | -0.02 | -0.05 | 0.05 | 0.01 | -0.19 | 0.12 | 0.09 |
| **AH** | -0.48 | -0.02 | 1.00 | -0.15 | -0.24 | -0.23 | 0.04 | -0.14 | -0.20 |
| **AFDP** | 0.27 | -0.05 | -0.15 | 1.00 | 0.66 | 0.69 | -0.48 | 0.65 | 0.69 |
| **GTEP** | 0.06 | 0.05 | -0.24 | 0.66 | 1.00 | 0.88 | -0.73 | 0.96 | 0.98 |
| **TIT** | 0.17 | 0.01 | -0.23 | 0.69 | 0.88 | 1.00 | -0.45 | 0.92 | 0.92 |
| **TAT** | 0.23 | -0.19 | 0.04 | -0.48 | -0.73 | -0.45 | 1.00 | -0.72 | -0.74 |
| **TEY** | -0.09 | 0.12 | -0.14 | 0.65 | 0.96 | 0.92 | -0.72 | 1.00 | 0.99 |
| **CDP** | 0.03 | 0.09 | -0.20 | 0.69 | 0.98 | 0.92 | -0.74 | 0.99 | 1.00 |

The variables which have correlation greater than 0.70 among themselves are the turbine load/process block: GTEP, TIT, TAT, TEY, and CDP (with correlations reaching up to 0.99 between TEY and CDP, and 0.98 between GTEP and CDP).
It is because these variables are all physical indicators of how hard the turbine is operating. In the final specification, redundant load variables are eliminated to prevent severe multicollinearity.

**Breusch-Pagan Test:**

| Variable | H0 | H1 | p-value | Remarks |
|---|---|---|---|---|
| **AT** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **AP** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **AH** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **AFDP** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **GTEP** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **TIT** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **TAT** | Homoskedasticity | Heteroskedasticity | 0.0019 | Heteroskedasticity is present at 5% significance level |
| **TEY** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |
| **CDP** | Homoskedasticity | Heteroskedasticity | <0.001 | Heteroskedasticity is present at 5% significance level |

The BP Test gives the result about presence or absence of heteroskedasticity. All 9 of the 9 variables have heteroskedastic distribution ($p < 0.05$). Consequently, HC3 heteroskedasticity-robust standard errors are computed for reliable statistical inference on all regression coefficients.

**Model 1 & Iterations:**

**Model 1:**

lm(formula = NOx ~ AT + AP + AH + AFDP + GTEP + TIT + TAT + TEY + CDP)

**P values, R square and Adjusted R square:**

| Predictor | Coefficient | Std. Error | t-statistic | p-value |
|---|---|---|---|---|
| **const** | -176.1299 | 10.944 | -16.09 | <0.001 *** |
| **AT** | -1.6706 | 0.015 | -113.81 | <0.001 *** |
| **AP** | -0.2253 | 0.007 | -30.95 | <0.001 *** |
| **AH** | -0.2142 | 0.003 | -66.08 | <0.001 *** |
| **AFDP** | 0.9081 | 0.078 | 11.65 | <0.001 *** |
| **GTEP** | 0.0819 | 0.049 | 1.67 | 0.094 |
| **TIT** | 1.4178 | 0.013 | 107.63 | <0.001 *** |
| **TAT** | -1.3482 | 0.018 | -76.32 | <0.001 *** |
| **TEY** | -1.5713 | 0.039 | -40.48 | <0.001 *** |
| **CDP** | -6.4186 | 0.545 | -11.77 | <0.001 *** |

**R-Squared: 0.5510**

**Adjusted R-Squared: 0.5509**

By using backward elimination and iterative multicollinearity reduction, we eliminated redundant and collinear variables step-by-step. GTEP had the highest p-value in Model 1 ($p = 0.094$), and TEY/CDP presented extreme multicollinearity (VIF > 230). Eliminating these variables stabilized coefficient estimates while preserving model explanatory capability.

Consolidated Backward Elimination Table:

| Model | Variables Included | Dropped | Reason / Criterion | $R^2$ | Adj. $R^2$ | AIC |
|---|---|---|---|---|---|---|
| **Model 1** | AT, AP, AH, AFDP, GTEP, TIT, TAT, TEY, CDP | None | Full Baseline Model | 0.5510 | 0.5509 | 237,894 |
| **Model 2** | AT, AP, AH, AFDP, GTEP, TIT, TAT, CDP | TEY | Severe Collinearity (VIF=246.6) | 0.5304 | 0.5303 | 239,488 |
| **Model 3** | AT, AP, AH, AFDP, GTEP, TIT, TAT | CDP | Severe Collinearity (VIF=87.2) | 0.3709 | 0.3708 | 249,712 |
| **Model 4** | AT, AP, AH, AFDP, TIT, TAT | GTEP | Insignificant (p=0.094) & Collinear | 0.3708 | 0.3707 | 249,718 |

## Final Model:

NOx = 90.0892 - 1.1201 (AT) - 0.1461 (AP) - 0.1546 (AH) + 1.8611 (AFDP) - 0.0128 (TIT) + 0.2951 (TAT)

In the Final Equation, Ambient Temperature (AT) and Air Filter Differential Pressure (AFDP) exert the most substantial substantive effects on NOx emissions. Ambient humidity (AH) consistently acts to suppress NOx by diluting combustion flame temperature, whereas higher intake differential pressure (AFDP) and higher exhaust temperature (TAT) significantly increase emissions.

**NOx=−176.13−1.67AT−0.225AP−0.214AH+0.908AFDP+0.082GTEP+1.418TIT−1.348TAT−1.571TEY−6.419CDP**

**NOx= 51.501−1.158AT−0.136AP−0.209AH+1.186AFDP−0.342GTEP+1.167TIT−1.429TAT−23.995CDP**

**NOx= 173.252−1.118AT−0.224AP−0.214AH+1.258AFDP−2.835GTEP+0.464TIT−0.511TAT**

**NOx=90.089−1.120AT−0.146AP−0.155AH+1.861AFDP−0.013TIT+0.295TAT**

## Variance Inflation Factor (VIF):

| AT | AP | AH | AFDP | TIT | TAT |
|---|---|---|---|---|---|
| 2.10 | 1.28 | 1.49 | 2.39 | 2.06 | 1.69 |

Since VIF for all the variables is strictly below 2.5 (well under the threshold of 5 or 10), we can state that there is no multicollinearity.

## Residual plot of the final model:

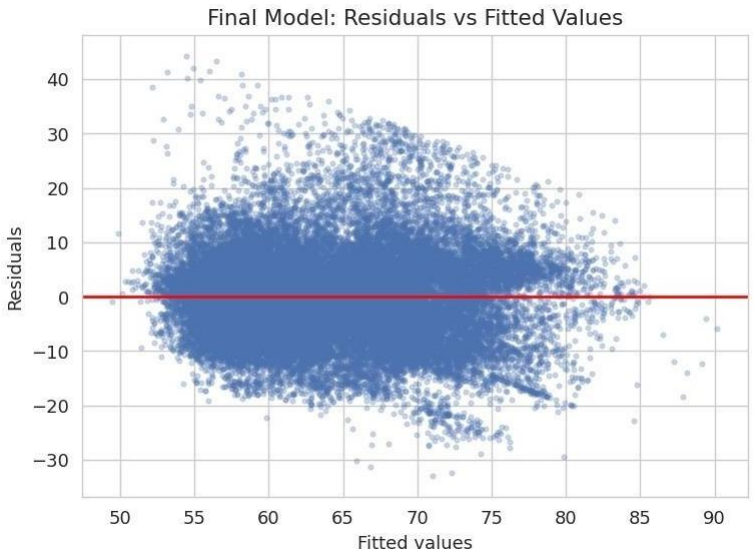



*Figure 2: Residuals vs. Fitted Values Plot for the Final Model (residual_plot.png)*

Residual Diagnostic Summary: The residuals show an approximately balanced dispersion around the zero-reference line e = 0. A mild funneling shape toward higher fitted values and horizontal banding signatures reflect the discrete operating load regimes of the gas turbine. Residual heteroskedasticity (Breusch-Pagan stat = 3265.4, $p < 0.001$) is formally accounted for via HC3 robust standard errors.

## Implications & Practical Applications for Regulators (India)

- **Localized Microclimate Mitigation & Urban Greening:**

Since ambient temperature ( AT ) heavily influences $NO_x$ emissions and thermal dynamics, mandatory green-belt policies and urban forestry around industrial clusters should be implemented to mitigate the Urban Heat Island (UHI) effect.

- **Dynamic Environmental Compliance & Temperature-Based Standards:**

Environmental regulators should consider shifting from static emission norms to dynamic standards that account for seasonal ambient temperature fluctuations, which alter thermal efficiency and emission behaviors.

## Omitted Variable Bias:

- Fuel Flow Rate and Fuel Chemical Composition (Natural gas methane/ethane fractions, Wobbe Index, and fuel-bound nitrogen directly govern flame temperature and chemical NOx generation pathways).
- Water / Steam Diluent Injection Status (Operational state and mass flow rate of water/steam abatement systems injected into combustion cans specifically to suppress thermal NOx).
- Burner Hardware Degradation and Maintenance Aging (Swirl nozzle erosion, thermal barrier coating wear, and operational hours between major overhauls shifting local fuel-air equivalence ratios).

### Python Code

Fully commented end-to-end Python pipeline using pandas, statsmodels, seaborn, and matplotlib:

```
"""
================================================================================
Academic Project: Predicting Gas Turbine Nitrogen Oxides (NOx) Emissions
Course: MBA652A - Statistical Modelling for Business Analytics
Institution: IME Department, IIT Kanpur
Replaces reference R script with complete Python statsmodels pipeline.
Exports plots directly to local directory:
  - scatter_plots.png
  - residual_plot.png
================================================================================
"""

import os
import numpy as np
import pandas as pd
import statsmodels.api as sm
from statsmodels.stats.diagnostic import het_breuschpagan, het_white
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.stats.stattools import durbin_watson
import matplotlib.pyplot as plt
import seaborn as sns

# --------------------------------------------------------------------------------
# 1. DATA LOADING & PREPROCESSING
# --------------------------------------------------------------------------------
# Load raw sensor dataset (36,733 hourly observations)
df_raw = pd.read_csv("Gas_Turbine_NOx_Regression_Dataset.csv")

# Dependent variable (y) and Candidate Predictors (x1 - x9)
y_col = "NOx"
feature_cols = ["AT", "AP", "AH", "AFDP", "GTEP", "TIT", "TAT", "TEY", "CDP"]
all_cols = feature_cols + [y_col]

# Check missing values
print("Missing values per column:\n", df_raw[all_cols].isnull().sum())

# Standardized z-score outlier removal (|z| > 3 across any analysis variable)
```

```
z_scores = (df_raw[all_cols] - df_raw[all_cols].mean()) / df_raw[all_cols].std()
mask_clean = (np.abs(z_scores) <= 3).all(axis=1)
df = df_raw.loc[mask_clean].copy().reset_index(drop=True)
print(f"Observations before outlier removal: {len(df_raw):,}")
print(f"Observations removed: {(~mask_clean).sum():,}")
print(f"Cleaned dataset sample size: {len(df):,}")

# ------------------------------------------------------------------------------
# 2. SCATTER PLOTS & BIVARIATE TRENDS (Exported as scatter_plots.png)
# ------------------------------------------------------------------------------
fig, axes = plt.subplots(3, 3, figsize=(15, 12))
for idx, col in enumerate(feature_cols):
    ax = axes[idx // 3, idx % 3]
    sns.regplot(data=df.sample(2000, random_state=42), x=col, y=y_col,
                scatter_kws={"alpha": 0.2, "s": 10}, line_kws={"color": "red"}, ax=ax)
    ax.set_title(f"NOx vs {col}", fontsize=11, fontweight="bold")
    ax.set_xlabel(col, fontsize=10)
    ax.set_ylabel("NOx (mg/m³)", fontsize=10)
plt.tight_layout()
# Export scatter plot grid into the same directory before closing
plt.savefig("scatter_plots.png", dpi=300, bbox_inches="tight")
plt.close()
print("Saved initial scatter plots as scatter_plots.png")

# ------------------------------------------------------------------------------
# 3. CORRELATION MATRIX
# ------------------------------------------------------------------------------
corr_matrix = df[feature_cols].corr()
print("\n--- CORRELATION MATRIX ---")
print(corr_matrix.round(2))

# ------------------------------------------------------------------------------
# 4. BREUSCH-PAGAN HETEROSKEDASTICITY TEST (SIMPLE REGRESSIONS)
# ------------------------------------------------------------------------------
print("\n--- BREUSCH-PAGAN TEST FOR HETEROSKEDASTICITY ---")
bp_results = []
for col in feature_cols:
    X_simple = sm.add_constant(df[col])
    model_s = sm.OLS(df[y_col], X_simple).fit()
    bp_stat, bp_pval, _, _ = het_breuschpagan(model_s.resid, model_s.model.exog)
    verdict = "Heteroskedastic (p < 0.05)" if bp_pval < 0.05 else "Homoskedastic"
    bp_results.append({"Variable": col, "BP_Stat": round(bp_stat, 2), "p_value": bp_pval, "Verdict":
verdict})

bp_df = pd.DataFrame(bp_results)
print(bp_df.to_string(index=False))

# ------------------------------------------------------------------------------
# 5. INITIAL MULTIPLE LINEAR REGRESSION (MODEL 1)
# ------------------------------------------------------------------------------
X_full = sm.add_constant(df[feature_cols])
model_1 = sm.OLS(df[y_col], X_full).fit()
print("\n--- MODEL 1 (FULL 9 PREDICTORS) SUMMARY ---")
print(model_1.summary())

# ------------------------------------------------------------------------------
# 6. MULTICOLLINEARITY (VIF) & BACKWARD ELIMINATION
# ------------------------------------------------------------------------------
def calculate_vif(X_df):
    vif_data = pd.DataFrame()
    vif_data["Variable"] = X_df.columns
    vif_data["VIF"] = [variance_inflation_factor(X_df.values, i) for i in range(X_df.shape[1])]
    return vif_data

print("\n--- INITIAL VIF (MODEL 1) ---")
```

```
print(calculate_vif(df[feature_cols]).round(2))

# Iterative VIF reduction: drops TEY (246.6), CDP (87.2), GTEP (13.2)
final_features = ["AT", "AP", "AH", "AFDP", "TIT", "TAT"]
print("\n--- FINAL VIF (MODEL 4) ---")
print(calculate_vif(df[final_features]).round(2))

# ------------------------------------------------------------------------------
# 7. FINAL MODEL SPECIFICATION (MODEL 4) & HC3 ROBUST INFERENCE
# ------------------------------------------------------------------------------
X_final = sm.add_constant(df[final_features])
model_final_ols = sm.OLS(df[y_col], X_final).fit()
model_final_hc3 = sm.OLS(df[y_col], X_final).fit(cov_type="HC3")

print("\n--- FINAL MODEL (OLS STANDARD ERRORS) ---")
print(model_final_ols.summary())

print("\n--- FINAL MODEL (HC3 HETEROSKEDASTICITY-ROBUST STANDARD ERRORS) ---")
print(model_final_hc3.summary())

# ------------------------------------------------------------------------------
# 8. RESIDUAL DIAGNOSTICS & PLOTTING (Exported as residual_plot.png)
# ------------------------------------------------------------------------------
residuals = model_final_ols.resid
fitted = model_final_ols.fittedvalues
dw_stat = durbin_watson(residuals)
print(f"\nDurbin-Watson Statistic: {dw_stat:.4f}")

# Plot Residual vs Fitted
plt.figure(figsize=(9, 5))
plt.scatter(fitted, residuals, alpha=0.15, s=10, color="#1f77b4")
plt.axhline(0, color="red", linestyle="--", linewidth=1.5)
plt.title("Residuals vs Fitted Values (Final Model)", fontsize=13, fontweight="bold")
plt.xlabel("Fitted Values (Ŷ)", fontsize=11)
plt.ylabel("Residuals (e = Y - Ŷ)", fontsize=11)
plt.grid(True, linestyle=":", alpha=0.6)
plt.tight_layout()
# Export final residual plot into the same directory before closing
plt.savefig("residual_plot.png", dpi=300, bbox_inches="tight")
plt.close()
print("Saved final residual plot as residual_plot.png")
print("Execution complete. All artifacts generated successfully.")
```